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Invited Review Article

Biomarkers in pediatric asthma: clinical maturity, intended use, and future directions

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ABSTRACT

Asthma in children is heterogeneous, and symptom-based assessment alone does not fully capture airway inflammation, future risk, or treatment responsiveness. A wide range of biomarkers has therefore been investigated, from clinically accessible inflammatory markers to emerging molecular and omics-based signatures. This narrative review, supported by a structured PubMed search, synthesizes pediatric asthma biomarker studies according to biomarker type, intended clinical use, feasibility in children, clinical maturity, and current limitations.

FeNO and blood eosinophil count are the most clinically mature biomarkers, particularly for identifying type 2 inflammation and supporting selected decisions regarding anti-inflammatory treatment or biologic therapy. IgE sensitization provides important phenotypic context for allergic asthma and anti-IgE treatment selection, whereas induced sputum offers direct airway inflammatory phenotyping but remains limited to specialist settings. Eosinophil granule proteins, especially eosinophil-derived neurotoxin, may reflect eosinophil activation rather than eosinophil number alone, but require further assay harmonization and pediatric reference ranges.

Other biomarkers, including periostin, cytokines, inflammatory proteins, microRNAs, breath-based markers, metabolomic and proteomic signatures, microbiome profiles, and multi-omics approaches, remain largely exploratory. Mechanistic studies highlight the importance of epithelial, immune, microbial, genetic, epigenetic, metabolic, and remodeling pathways in pediatric asthma endotyping. However, most emerging biomarkers are not yet clinically actionable.

Future biomarker development should focus on clearly defined intended uses, age-appropriate validation, standardized assays, reproducible cutoffs, and evidence that biomarker-guided decisions improve outcomes beyond conventional clinical assessment, lung function, FeNO, blood eosinophils, and allergen sensitization.

Introduction

Asthma is one of the most common chronic diseases in childhood and is characterized by marked heterogeneity in clinical presentation, airway inflammation, lung function impairment, and treatment response.[1–3] In children, symptom-based assessment alone may not adequately capture abnormal lung function, active airway inflammation, or future exacerbation risk.[4] Objective measurements such as spirometry and fractional exhaled nitric oxide (FeNO) can identify clinically relevant abnormalities even in children who report good symptom control. This is clinically important because unrecognized or insufficiently controlled airway inflammation, together with impaired lung growth and other pathophysiological abnormalities, may influence respiratory health across the life course, contributing to persistent asthma, reduced maximally attained lung function, accelerated lung function decline, and severe respiratory outcomes in adulthood[5,6,7] Therefore, early and appropriate assessment of disease activity, inflammatory phenotype, and future risk is essential not only for short-term symptom control, but also for preserving lifelong lung health and potentially improving long-term prognosis.

Biomarkers have therefore attracted increasing attention as tools to complement conventional clinical assessment in pediatric asthma.[8] Rather than serving as stand-alone diagnostic tests, biomarkers may support several distinct clinical purposes, including identification of type 2 airway inflammation, prediction

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of corticosteroid responsiveness, assessment of exacerbation risk, selection of biologic therapies, monitoring of treatment response, and evaluation of treatment adherence. FeNO, blood eosinophil count, total and allergen-specific IgE, allergen sensitization, induced sputum inflammatory phenotypes, eosinophil activation markers, periostin, cytokines, breath-based markers, miRNAs, and multi-omics signatures have all been investigated in pediatric populations.[2,9]

However, important gaps remain in the clinical interpretation of pediatric asthma biomarkers. Numerous biomarkers have been investigated, but they differ markedly in clinical maturity, feasibility, standardization, and actionability. In addition, their intended use is often not clearly distinguished: a marker useful for identifying type 2 inflammation may not be equally useful for diagnosis, prediction of exacerbations, biologic selection, treatment monitoring, or adherence assessment. This distinction is particularly important in children, because developmental differences, airway growth, age-dependent reference values, atopic comorbidities, viral triggers, and practical limitations in sample collection may alter the interpretation and utility of biomarkers compared with adults.

This review aims to provide a clinically oriented synthesis of pediatric asthma biomarkers according to their clinical maturity, intended use, feasibility in children, and major limitations.

Literature search and narrative synthesis

To provide a transparent evidence base for this narrative review, we conducted a structured PubMed search of pediatric asthma biomarker studies. The search was performed on May 31, 2026, using two complementary strategies: one targeting established clinical biomarkers and the other targeting emerging mechanistic and omics-based biomarkers. Full search queries, eligibility criteria, and categorization procedures are provided in the Supplementary Methods.

Records were exported in NBIB format and screened using Rayyan (Rayyan Systems Inc., Cambridge, MA, USA). Eligible studies included children or adolescents with asthma and evaluated biomarkers in relation to asthma-related clinical outcomes, including disease severity, exacerbations, asthma control, treatment response, prognosis, phenotypes, or endotypes. Reviews, editorials, protocols, conference abstracts, case reports, adult-only studies, animal or in vitro studies without human clinical data, retracted articles, non-English publications, and studies without asthma-related clinical outcomes were excluded.

After screening, 414 studies were retained and categorized according to the primary role of the biomarker evidence: core clinical biomarker studies, supportive clinical studies, and mechanistic biomarker studies. Core clinical biomarker studies were those in which biomarkers were a primary focus and directly linked to clinically relevant pediatric asthma outcomes. Supportive studies provided contextual or preliminary clinical evidence, whereas mechanistic studies primarily investigated molecular pathways, biological endotypes, or omics-based signatures. The study selection and categorization process is summarized in Figure 1. Because of substantial heterogeneity in biomarker types, study designs, outcomes, and analytical methods, findings were synthesized narratively rather than by meta-analysis.

Core clinical biomarkers

Overview

Among the 125 core clinical biomarker studies, each article was assigned to one primary biomarker category based on its main biomarker or dominant analytical focus. This classification provides the framework for Table 1, which summarizes the major biomarker groups, their principal clinical signal or use, and their current level of clinical maturity. Because some studies evaluated more than one biomarker, study counts should be interpreted as a descriptive overview of research activity rather than as a measure of evidence quality or clinical utility.

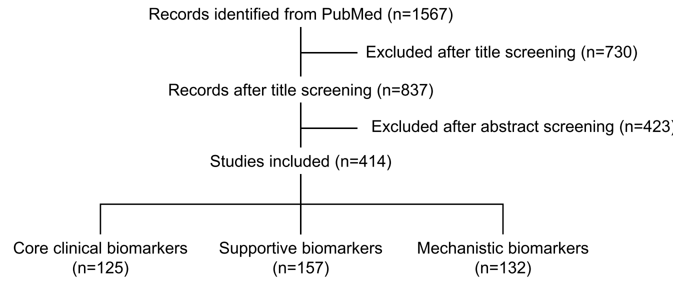

Fig. 1. Flow diagram of study selection and biomarker categorization.
A total of 1567 records were identified from PubMed. After title and abstract screening, 414 studies were included and categorized as core clinical biomarker studies, supportive biomarker studies, or mechanistic biomarker studies.

FeNO was the most frequently investigated biomarker, accounting for approximately half of the core clinical studies, followed by blood eosinophil count. Other categories included IgE or allergen sensitization, induced sputum inflammatory phenotypes, physiological markers such as lung function, forced oscillation technique, or bronchial responsiveness, eosinophil granule proteins, and a heterogeneous group of emerging or exploratory biomarkers, including periostin, cytokines and inflammatory proteins, breath-based markers, miRNAs, epigenetic or genetic markers, microbiome-related markers, and multi-omics approaches.

To avoid overinterpreting study counts alone, each biomarker group was also categorized according to clinical maturity and actionability. This assessment considered consistency of pediatric evidence, feasibility of measurement in children, assay reproducibility, interpretability in relation to age, growth, atopy, comorbid allergic disease, and treatment status, evidence of added value beyond conventional clinical assessment, and potential to inform selected clinical decisions. Based on these criteria, biomarker groups were classified as clinically mature biomarkers, clinical phenotype markers, specialist-use biomarkers, clinical physiological markers, investigational clinical biomarkers, or exploratory biomarkers.

Fractional exhaled nitric oxide (FeNO)

FeNO is the most clinically mature non-invasive biomarker in pediatric asthma. Mechanistically, FeNO reflects nitric oxide production in the airway epithelium, largely through inducible nitric oxide synthase. Increased nitric oxide synthesis in asthma is supported by evidence of increased epithelial NOSII expression, transcriptional activation, and corticosteroid-sensitive reduction of NOSII expression in asthmatic airways.[10] Type 2 cytokines, particularly IL-13, further enhance iNOS expression and nitrite production in primary human bronchial epithelial cells, providing a biological link between type 2 airway inflammation and elevated FeNO.[11] Thus, FeNO is best understood as a marker of type 2/eosinophilic airway inflammation rather than as a marker of asthma itself.

This distinction is central to its clinical interpretation. The 2011 ATS guideline recommends FeNO for identifying eosinophilic airway inflammation and estimating the likelihood of

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Table 1
Overview and clinical maturity of core clinical biomarker groups in pediatric asthma.

Biomarker group	n	Main clinical signal or use	Current status
FeNO	62	Non-invasive marker of type 2, eosinophilic airway inflammation; useful for anti-inflammatory treatment monitoring and selected risk assessment	Clinically mature biomarker
Blood eosinophil count	18	Systemic type 2, eosinophilic inflammation; useful for risk stratification and biologic eligibility	Clinically mature biomarker
IgE and allergen sensitization	4	Allergic asthma phenotype, sensitization profile, and anti-IgE treatment eligibility	Clinical phenotype marker
Induced sputum inflammatory phenotypes	4	Direct assessment of airway inflammatory phenotype, including eosinophilic and neutrophilic patterns	Specialist-use biomarker
Lung function, FOT, and bronchial responsiveness	7	Physiological abnormalities, airway obstruction, small airway dysfunction, or airway hyperresponsiveness	Clinical physiological marker
Eosinophil granule proteins	9	Eosinophil activation markers, including ECP, EDN (EPX), and EPO	Investigational clinical biomarker
Other emerging or exploratory biomarkers	21	Periostin, cytokines/inflammatory proteins, breath-based markers, miRNA, epigenetic/genetic markers, microbiome, and multi-omics signatures	Exploratory biomarker
Total	125		

Each study was assigned to one primary biomarker category according to the main biomarker or dominant analytical focus. Counts describe the distribution of the core clinical biomarker literature and should not be interpreted as a formal measure of evidence quality or clinical utility. Current status was assigned qualitatively according to clinical maturity and actionability, not by study count alone.

“Clinically mature biomarker” indicates a biomarker with relatively standardized measurement, substantial clinical experience, and an established interpretive framework for selected clinical questions, but not necessarily stand-alone diagnostic or therapeutic utility. “Clinical phenotype marker” indicates a marker useful primarily for defining allergic or inflammatory phenotype rather than monitoring disease activity. “Specialist-use biomarker” indicates a clinically relevant marker whose routine use is limited by feasibility, invasiveness, sample processing, or need for specialized expertise. “Clinical physiological marker” indicates an objective physiological measure relevant to asthma diagnosis or risk assessment, but not a direct inflammatory biomarker. “Investigational clinical biomarker” indicates a biomarker evaluated in pediatric clinical studies but requiring further validation, assay harmonization, reference ranges, or evidence of incremental clinical value before routine implementation. “Exploratory biomarker” indicates a marker mainly used for mechanistic insight, endotype discovery, or hypothesis generation.

FeNO, fractional exhaled nitric oxide; FOT, forced oscillation technique; ECP, eosinophil cationic protein; EDN, eosinophil-derived neurotoxin; EPX, eosinophil protein X; EPO, eosinophil peroxidase; VOCs, volatile organic compounds.

corticosteroid responsiveness, and suggests that FeNO may support the diagnosis of asthma when objective evidence is needed. However, it also emphasizes that asthma is a clinical diagnosis and that FeNO is not a stand-alone diagnostic test.[12] In children, the ATS guideline proposed clinically useful interpretive ranges: FeNO below 20 ppb makes eosinophilic inflammation and corticosteroid responsiveness less likely, FeNO above 35 ppb makes these more likely in symptomatic children, and intermediate values require cautious interpretation in clinical context.[12] The ERS pediatric diagnostic guideline also recommends FeNO as part of the diagnostic work-up in children aged 5–16 years with suspected asthma, while noting that FeNO ≥25 ppb is supportive of asthma and FeNO <25 ppb does not exclude asthma.[13]

The diagnostic role of FeNO is therefore supportive, not definitive. A pediatric systematic review and meta-analysis found moderate diagnostic performance for childhood asthma, with pooled sensitivity 0.79, specificity 0.81, and SROC 0.87, but substantial heterogeneity across studies.[14] In a school-age clinical cohort, FeNO performed better than spirometry and similarly to induced sputum eosinophils for distinguishing steroid-naïve asthma from non-asthma, with an optimal cutoff around 19 ppb.[15] These findings support FeNO as an objective inflammatory marker that can strengthen diagnostic confidence, particularly in steroid-naïve, type 2–high, or allergic asthma, but they do not justify using FeNO alone to diagnose or exclude asthma.

FeNO also has potential roles in monitoring airway inflammation, assessing corticosteroid responsiveness, and identifying possible non-adherence. In children with persistent asthma, FeNO decreased rapidly during inhaled budesonide treatment, rose after steroid withdrawal, and decreased again in the group randomized back to inhaled steroid therapy, whereas conventional lung function variables did not clearly distinguish treated from untreated children. Importantly, FeNO reduction correlated with measured adherence to budesonide, suggesting a role in detecting poor adherence or inadequate anti-inflammatory exposure.[16] The ATS guideline similarly lists evaluation of adherence to anti-inflammatory therapy among the common reasons for measuring FeNO.[17] In longitudinal pediatric asthma, repeated FeNO measurements may be more informative than isolated values: in atopic children, the highest FeNO value and frequency of FeNO elevation over serial monitoring independently predicted later loss of asthma control.[18] In the biologic era, FeNO has gained additional relevance as a type 2 biomarker for risk stratification and treatment selection. In children aged 6–11 years with uncontrolled moderate-to-severe asthma in the LIBERTY ASTHMA VOYAGE trial, higher baseline FeNO and blood eosinophil counts were associated with higher exacerbation risk and greater clinical response to dupilumab. Interaction analyses suggested that FeNO predicted exacerbation response independently of blood eosinophil count.[19] This supports FeNO as a useful marker of IL-4/IL-13–driven airway inflammation in children being considered for type 2–targeted therapy.

However, FeNO interpretation in children has important limitations. FeNO is strongly influenced by allergic sensitization, allergen exposure, age, rhinitis, smoking exposure, and corticosteroid treatment. In the COPSAC cohorts, asthma was associated with higher FeNO only among children with aeroallergen sensitization, and asthma-associated traits such as blood eosinophils, total IgE, bronchodilator response, and bronchial hyperresponsiveness were linked to FeNO mainly in sensitized children.[20] Similarly, in the BAMSE adolescent cohort, the upper limit of normal FeNO depended strongly on the presence, type, and degree of allergic sensitization; sensitization-specific cutoffs differed substantially from fixed cutoffs.[21] Therefore, an elevated FeNO in an atopic child may indicate type 2 airway inflammation, but it may also reflect allergen exposure or upper airway allergic disease rather than uncontrolled lower airway asthma.

Finally, FeNO-guided treatment algorithms have shown mixed clinical benefit. The 2021 ATS guideline made only a conditional recommendation for FeNO-based care in patients with asthma in whom treatment is being considered, based on overall low-quality evidence.[8] Randomized trials in children have not consistently shown that routine FeNO-guided adjustment improves exacerbations, lung function, asthma control, or quality of life.[22,23] Thus, FeNO should be used as an adjunctive biomarker to answer specific clinical questions: Is type 2/eosinophilic inflammation likely? Is corticosteroid responsiveness likely? Is poor adherence possible? Is persistent inflammation present despite apparent

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clinical control? Is the child likely to benefit from type 2–targeted therapy? It should not be used as a stand-alone diagnostic test or as an automatic trigger for treatment escalation.

In summary, FeNO is a clinically mature, non-invasive marker of type 2/eosinophilic airway inflammation and corticosteroid responsiveness. Its interpretation should remain context-based, incorporating symptoms, lung function, allergic sensitization, rhinitis, allergen exposure, corticosteroid use, and the intended clinical question (Fig.2).

Blood eosinophil count

Blood eosinophil count is a readily available biomarker in pediatric asthma. Its major strengths are low cost, feasibility across age groups, and applicability even in young children in whom lung function testing or FeNO measurement may be difficult. Eosinophil development and survival are largely regulated by IL-5,[24] and blood eosinophil count may therefore represent the "IL-5–related" component of type 2 inflammation, in contrast to FeNO, which more directly reflects airway epithelial IL-4/IL-13 activity. Pediatric studies support its clinical relevance, particularly for exacerbation risk. In preschool children with recurrent wheezing, prespecified blood eosinophil cut points from ≥150 to ≥350 cells/μL were associated with higher odds of exacerbation, higher exacerbation rates, and greater hospitalization occurrence.[25] However, specificity was poor, and prediction improved when eosinophil counts were combined with allergic sensitization markers, indicating that blood eosinophils are more useful as part of a broader type 2 profile than as a stand-alone predictor. In

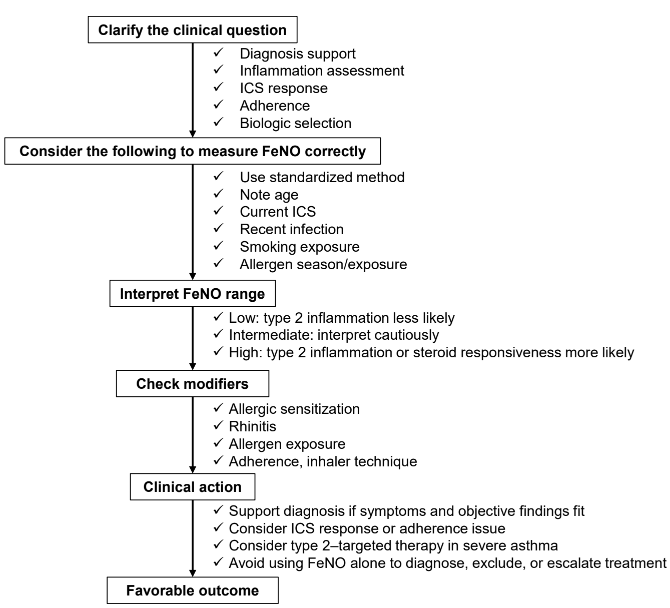

Fig. 2. Context-based interpretation of FeNO in children with asthma or suspected asthma.
Before measuring FeNO, the intended clinical question should be clarified, such as diagnostic support, assessment of type 2 airway inflammation, prediction of inhaled corticosteroid responsiveness, evaluation of adherence, or biologic treatment selection. FeNO should then be measured using a standardized method, with attention to factors that may affect accuracy and stability, including age, current inhaled corticosteroid use, recent respiratory infection, smoking exposure, and allergen season or exposure. FeNO values should be interpreted as low, intermediate, or high, and the interpretation should be further modified by allergic sensitization, rhinitis, allergen exposure, adherence, and inhaler technique. Clinical action should be based on this context-based interpretation rather than on the FeNO value alone, with the goal of supporting appropriate diagnosis, anti-inflammatory treatment decisions, adherence assessment, biologic selection, and ultimately better clinical outcomes.

moderate-to-severe pediatric asthma, the interpretation of blood eosinophilia is also age-dependent. In the SAMP cohort, third-quartile thresholds differed between preschool and school-age children, and high eosinophil counts were linked to atopic features and hospitalization history in younger children, but to staphylococcal toxin–specific IgE sensitization in school-age children.[26]

Blood eosinophils also contribute to treatment selection in more severe disease. In children aged 6–11 years with uncontrolled moderate-to-severe asthma in the LIBERTY ASTHMA VOYAGE trial, higher baseline blood eosinophil counts and FeNO were associated with greater exacerbation risk and better response to dupilumab.[19] Consistent with this, current severe asthma guidance uses blood eosinophils as one marker of type 2 inflammation and as part of biologic eligibility or response prediction, while emphasizing that biomarker levels may fluctuate and can be suppressed by corticosteroids.[27]

The main limitation is lack of airway specificity. Blood eosinophil count can be influenced by age, atopy, allergic comorbidities, infection, corticosteroid exposure, biologic therapy, and non-asthma eosinophilic disorders.[28,29] It does not reliably diagnose asthma, define asthma control, or prove airway eosinophilia. Its clinical value is greatest when combined with FeNO, allergic sensitization, exacerbation history, lung function, and treatment exposure. Simultaneous elevation of FeNO and blood eosinophils has been associated with greater asthma morbidity than either marker alone, supporting their complementary roles as local and systemic type 2 biomarkers.[30]

In clinical practice, blood eosinophil count is best used to identify a systemic type 2/eosinophilic phenotype, estimate exacerbation risk, and support biologic selection. It should not be used as a stand-alone diagnostic or treatment-escalation marker, but as part of an integrated assessment together with FeNO, sensitization, symptoms, exacerbation history, and lung function.

IgE and allergen sensitization

Total IgE, allergen-specific IgE, and allergen sensitization are commonly assessed in children with asthma, but their biomarker role differs from markers of current airway inflammation. Their clinical relevance is best understood in relation to allergic asthma phenotypes, interpretation of type 2 inflammation, and selection of allergen- or IgE-targeted treatment.

Sensitization has been linked to childhood wheezing phenotypes. In the Southampton Women's Survey, longitudinal wheeze phenotypes in the first 6 years of life were differentiated by both lung function and atopic sensitization. Persistent and late-onset wheeze were associated with atopy at 3 and 6 years, intermediate-onset wheeze showed an earlier association with atopy at 1 year, and these wheezing phenotypes were also associated with FeNO at 6 years.[31] Allergen sensitization is also closely related to type 2 biomarker expression in established childhood asthma. In the BAMSE birth cohort, sensitized adolescents with asthma had higher FeNO and blood eosinophil levels than non-sensitized asthmatic adolescents, and allergic asthma was associated with peripheral airway dysfunction measured by impulse oscillometry. In contrast, non-sensitized asthma showed airflow obstruction without the same pattern of eosinophilic inflammation or small-airway involvement.[32]

Sensitization burden may further inform anti-IgE treatment selection. In a pooled analysis of two randomized trials of omalizumab in inner-city children and adolescents with persistent asthma, greater aeroallergen sensitization burden, higher total IgE, and higher eosinophil counts were associated with greater fall exacerbation risk in placebo-treated participants and greater benefit from omalizumab. Multiple aeroallergen sensitization was the strongest predictor of response; children sensitized to four or more aeroallergen groups had a 51% reduction in the odds of fall exacerbation with omalizumab.[33] In the present review, however, we did not identify consistent pediatric evidence that total IgE, allergen-specific IgE, or sensitization burden directly reflects asthma severity or current asthma control. Total IgE is influenced by age, atopic dermatitis, polysensitization, and other allergic or eosinophilic conditions, while allergen-specific IgE or skin testing demonstrates sensitization rather than clinically relevant allergen-driven asthma. Therefore, these markers should be interpreted together with allergen exposure, seasonality, rhinitis, symptoms, exacerbation history, FeNO, blood eosinophils, lung function, and treatment response.

In clinical practice, IgE and allergen sensitization are best regarded as clinical phenotype markers rather than direct markers of asthma activity. They help identify allergic asthma, contextualize FeNO and blood eosinophil findings, guide allergen-focused management, and support anti-IgE treatment selection.

Induced sputum inflammatory phenotypes

Induced sputum analysis provides a relatively direct assessment of airway inflammatory cells and can classify asthma or recurrent wheezing into eosinophilic, neutrophilic, mixed granulocytic, and paucigranulocytic inflammatory phenotypes. Its main value in children is detailed inflammatory phenotyping in selected cases, rather than routine monitoring. Pediatric studies support the biological and clinical relevance of sputum eosinophilia. In children aged 5–10 years with newly diagnosed mild persistent asthma, sputum eosinophils were higher than in healthy controls, correlated with bronchial hyperresponsiveness, and decreased after budesonide treatment together with improvement in symptoms and lung function. The response was most evident in children with higher baseline sputum eosinophilia.[34] In younger children and infants with recurrent wheezing, induced sputum cell profiles also identified heterogeneous inflammatory phenotypes; neutrophilic inflammation was most common overall, whereas sputum eosinophils were more frequent in severe wheezing and increased across mild, moderate, and severe groups.[35]

Sputum eosinophilia may also have prognostic relevance. A childhood asthma cohort classified by induced sputum at inception showed that eosinophilic airway inflammation was associated with more persistent wheeze during follow-up, suggesting that sputum eosinophilia can identify a clinically meaningful inflammatory phenotype.[36] However, sputum inflammatory phenotypes are not necessarily stable over time in children with asthma, which limits the interpretation of a single sputum result as a fixed disease trait.[37]

The main limitations are feasibility and scalability. Sputum induction requires cooperation, trained staff, standardized processing, and adequate sample quality. Although it can be performed safely in school-aged children and has also been studied in younger wheezing children, it remains more invasive, time-consuming, and less widely available than FeNO or blood eosinophil count. In clinical practice, induced sputum is therefore best regarded as a specialist-use biomarker for airway inflammatory phenotyping and corticosteroid-responsive eosinophilic inflammation, particularly in difficult, severe, or discordant cases.

Eosinophil granule proteins

Eosinophil granule proteins, including eosinophil cationic protein (ECP), eosinophil-derived neurotoxin (EDN, also known as eosinophil protein X), eosinophil peroxidase, and major basic protein, are released during eosinophil activation and may therefore reflect eosinophil secretory activity rather than eosinophil number alone. This distinction is clinically relevant because blood eosinophil count indicates the size of the circulating eosinophil pool, whereas granule proteins may better capture eosinophil activation, degranulation, and turnover. A recent review emphasized that eosinophil-derived proteins can provide a more complete picture of eosinophil activation status than eosinophil counts alone.[38]

Among these proteins, ECP has been studied for many years in pediatric asthma. Serum ECP has been associated with recent symptoms and subsequent clinical course in children with asthma, suggesting potential value for monitoring eosinophilic inflammation.[39] In children with asthma, serum EDN has been associated with atopic asthma, asthma severity, and bronchial hyperresponsiveness,[40,41] and in some studies showed clearer differences across severity groups than ECP.[40] More recent pediatric studies also support EDN as a marker of eosinophilic activity: urinary EDN/creatinine correlated with blood eosinophils, type2-high phenotype, and reduced lung function in the ALLIANCE cohort, while serum EDN was associated with current asthma, acute wheeze severity, recovery after treatment, and IgE sensitization in preschool children.[42,43]

An important practical difference among granule proteins relates to their physicochemical properties. ECP is strongly cationic and relatively "sticky," which can promote nonspecific binding to storage tubes and contribute to pre-analytical variability. EDN is less cationic than ECP, with a lower isoelectric point, and has shown negligible binding to polypropylene tubes. EDN also appears stable under clinically relevant storage conditions, including room-temperature whole blood storage and repeated freeze–thaw cycles. These features may improve analytical reproducibility and make EDN more suitable than ECP for standardized clinical or research assays.[38]

Overall, serum or urinary measurements of eosinophil granule proteins may provide information complementary to blood eosinophil count, particularly in relation to eosinophil activation, disease severity, lung function impairment, and treatment response. At present, however, further validation, assay harmonization, and pediatric reference ranges with clinically actionable cutoffs are needed before routine implementation.

Emerging and exploratory biomarkers

The final category in Table 1 comprises biomarkers evaluated in core clinical studies but not yet sufficiently mature for routine pediatric use. Because this category overlaps with supportive clinical studies of similar candidate markers, core exploratory and relevant supportive studies are considered together in this section. These biomarkers include epithelial and remodeling-related proteins, cytokines and inflammatory proteins, genetic, epigenetic, and microRNA markers, breath-based markers, metabolomic and proteomic signatures, and microbiome- or transcriptome-based profiles (Table 2). Their current value lies mainly in biological insight and phenotype refinement rather than routine clinical decision-making.

Periostin

Periostin is biologically attractive because it is induced by type 2 cytokine signaling and may reflect epithelial–mesenchymal activation.[44,45] In preschool children with recurrent wheezing, higher serum periostin predicted wheezing exacerbations during the following year, particularly in atopic wheeze.[46] However,

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interpretation in children is complicated by developmental variation: serum periostin is higher in children than in adults, changes with age and growth, and may be influenced by bone turnover and extracellular matrix remodeling.[47,48] In the DADO study of children with poorly controlled asthma, serum periostin levels were high overall and tended to increase with age. However, they were not correlated with IgE, eosinophils, FEV1, or FeNO, and their relationship with uncontrolled severe asthma was not straightforward.[49,50] Thus, periostin remains a biologically relevant but exploratory pediatric biomarker.

YKL-40

YKL-40 has been proposed as a remodeling-related biomarker in adult asthma, but pediatric studies have not consistently demonstrated clinically useful associations, supporting its current classification as exploratory in children.[51,52]

Cytokines and inflammatory proteins

Cytokines and inflammatory proteins may provide pathway-specific information beyond conventional type 2 biomarkers, although pediatric evidence remains limited. IL-26 has been evaluated in two pediatric studies with somewhat different implications: local airway IL-26 was proposed as a marker related to disease severity in pediatric asthma without clear type 2 inflammatory features, whereas systemic IL-26 was later reported to correlate with better asthma control in children sensitized to dog allergen.[53,54] HMGB1 has also been investigated as a marker linking airway epithelial injury, repair, and inflammatory amplification. Experimental studies suggest that HMGB1 can promote extracellular matrix synthesis and wound repair in bronchial epithelial cells[55] and may contribute to type 2 responses in the later stage of respiratory syncytial virus infection.[56] In pediatric asthma, sputum HMGB1 was reported to reflect disease status, and serum HMGB1 was explored as a biomarker of inhaled corticosteroid treatment response in children with moderate-to-severe asthma.[57] These findings suggest that soluble inflammatory mediators may capture noncanonical inflammatory or epithelial-injury pathways, but they remain exploratory because of limited replication, heterogeneous sampling compartments, and uncertain clinical actionability.

Breath-based markers

Breath-based biomarkers, including volatile organic compounds, electronic-nose profiles, and exhaled breath condensate markers, are attractive because they are non-invasive and potentially suitable for repeated assessment in children. Exhaled volatile organic compound profiles predicted exacerbations in a 1-year prospective study of children with asthma, and a subsequent study suggested that VOC patterns may predict exacerbations within 14 days after sampling.[58,59] Electronic-nose systems use sensor arrays to recognize overall breath-pattern signatures of volatile compounds rather than quantifying individual molecules, and electronic-nose profiling has also shown potential for discriminating uncontrolled asthma in children.[60] However, breath-based biomarkers remain highly dependent on sampling conditions, devices, analytical pipelines, and external validation, and are therefore best regarded as exploratory tools.

Taken together, the biomarkers summarized in Table 2 remain exploratory or supportive rather than routinely actionable. They provide biological and phenotypic insight, but their clinical use will depend on further validation and demonstration of incremental value beyond currently available clinical and inflammatory markers.

Mechanistic biomarkers and future directions

The clinical biomarker categories summarized in Tables 1 and 2 were based primarily on studies evaluating biomarkers in relation to clinically interpretable pediatric asthma domains, such as severity, exacerbations, control, phenotype, endotype, prognosis, or treatment response. In contrast, mechanistic biomarker studies were analyzed separately because their main purpose was to clarify disease biology rather than to establish clinically actionable biomarker tools. To maintain consistency with the preceding tables, these studies were summarized by dominant biological domain rather than listed individually (Table 3).

Several studies have highlighted the airway epithelium as a central site of biomarker discovery. Nasal and airway transcriptomic studies have identified type 2 epithelial response modules, mucus-related pathways, interferon responses, epithelial injury signatures, and ciliary-function networks that may distinguish pediatric asthma phenotypes or relate to lung function and exacerbation risk.[61] In high-risk urban children, an allergic asthma endotype with airway obstruction was linked to epithelial IL-13 and MUC5AC-related modules.[62] Network analysis of nasal transcriptomes also identified master regulator genes and modules related to ciliary function and inflammatory responses in children with severe asthma.[62,63]

Genetic, epigenetic, and regulatory RNA studies formed the largest mechanistic domain. These studies included DNA methylation, genetic–epigenetic interactions, GWAS/eQTL-based analyses, microRNAs, long non-coding RNAs, and other post-transcriptional regulatory markers. DNA methylation studies in airway epithelial cells, peripheral blood, and inner-city pediatric cohorts have identified asthma-associated epigenetic signatures, suggesting that environmental exposure and immune development may leave measurable molecular imprints related to asthma susceptibility.[64] Rhinovirus infection has also been shown to induce DNA methylation and mRNA-expression changes in nasal

Table 2
Emerging and exploratory biomarker groups in pediatric asthma.

Biomarker group	Examples	Potential contribution
Epithelial/remodeling-related proteins	Periostin, YKL-40	Type 2-associated tissue response; remodeling biology; wheeze/exacerbation risk research
Cytokines/inflammatory proteins	IL-26, HMGB1, soluble immune mediators	Pathway-specific inflammation; severe or non–type 2 asthma biology
Genetic/epigenetic/miRNA markers	miR-21, EBC miRNAs, DNA methylation markers	Molecular endotyping; treatment-response research
Breath-based markers	VOCs, eNose, exhaled breath condensate markers	Non-invasive exacerbation or control prediction
Metabolomics/proteomics	Metabolic or protein signatures	Severity biology; pathway discovery
Microbiome/transcriptome/multi-omics	Airway microbiome, host transcriptome, integrated signatures	Exacerbation biology; endotype discovery

Biomarker groups listed here correspond to the “other emerging or exploratory biomarkers” category in Table 1 and related supportive clinical studies. They may provide mechanistic or phenotypic insight but are not yet sufficiently standardized or validated for routine pediatric asthma management.

Table 3
Mechanistic biomarker studies by biological domain.

Biological domain	n	Main focus
Genetic, epigenetic, and regulatory RNA biomarkers	48	DNA methylation, genetic–epigenetic interactions, GWAS/eQTL, miRNA, lncRNA, piRNA, post-transcriptional regulation
Epithelial/airway transcriptomics and single-cell profiling	25	Nasal or airway transcriptomics, epithelial type 2 modules, interferon responses, mucus pathways, single-cell immune or epithelial profiles
Immune-cell, cytokine, and inflammatory pathway profiling	22	T-cell, eosinophil, neutrophil, dendritic-cell, mast-cell, cytokine, and innate immune pathway signatures
Metabolomics, lipidomics, proteomics, and exhaled/biofluid signatures	14	Metabolomic, lipidomic, proteomic, exhaled-breath, bronchoalveolar lavage, blood, saliva, or urine signatures
Microbiome, virome, and host–microbe interactions	12	Airway, nasopharyngeal, or gut microbiome; virome; metagenomic or metatranscriptomic host–microbe interactions
Integrative multi-omics and computational endotyping	9	Multi-omics integration, network-based analysis, machine-learning phenotypes, computational risk/endotype discovery
Remodeling and extracellular matrix pathways	2	Airway remodeling, extracellular matrix organization, lumican, pro-remodeling epithelial factors
Total	132	

Studies were assigned to one dominant biological domain according to the main platform, pathway, or analytical focus of the study. Counts are descriptive and are intended to show the distribution of mechanistic biomarker research identified in the systematic search. Mechanistic studies were used to inform disease biology, endotype discovery, and future biomarker development, rather than to define currently actionable clinical biomarker categories.

epithelial cells from children with asthma, linking viral exposure to epithelial epigenetic regulation.[65] Regulatory RNA studies, including circulating microRNA profiling, have suggested potential roles in exacerbation risk and treatment response in childhood asthma.[66,67] Together, these studies are valuable for identifying pathways through which early-life exposures, viral infection, allergen sensitization, and immune development may influence asthma inception or persistence; however, most remain discovery-oriented and require replication across age groups, tissues, and populations.

Microbiome, virome, and host–microbe studies further illustrate the importance of airway ecology in pediatric asthma. The upper-airway microbiota has been associated with loss of asthma control and exacerbation risk in children,[68] and seasonal airway microbiome–transcriptome interactions have been linked to asthma exacerbation susceptibility.[69] These findings suggest that microbial composition and host immune responses may jointly influence periods of increased risk.

Metabolomic, lipidomic, proteomic, and multi-omics studies have expanded pediatric asthma biomarker research from individual molecules to integrated molecular signatures. Salivary proteomic profiling has identified candidate protein signatures in children with asthma, and metabolomic–immunological profiling has suggested lipid, amino-acid, and immune-pathway differences according to asthma severity.[70,71] Deep multi-omic profiling of lower-airway samples further identified molecular factors characterizing preschool recurrent wheeze and an inferred trajectory from health to wheeze and school-age asthma.[72] These approaches may help define biologically meaningful endotypes that are not captured by symptoms, lung function, or single type 2 biomarkers alone, but their current value remains primarily in pathway discovery and endotype refinement.

Remodeling and extracellular matrix pathways represent a smaller but clinically important mechanistic domain. Airway remodeling can be present early in preschool children with severe recurrent wheeze,[73] and epithelial cell-derived lumican has been reported to modulate extracellular matrix dynamics by altering fibroblast function and collagen-related remodeling in early-life airway disease.[74] Such studies illustrate how mechanistic biomarkers may connect epithelial signals with structural airway changes relevant to persistent wheeze and asthma.

At present, mechanistic biomarkers should not be interpreted as clinically actionable tests. Their main contribution is to identify biological pathways, generate hypotheses, and define candidate endotypes for future validation. These studies support a gradual shift from single-marker assessment toward integrated pediatric asthma endotyping.

Conclusion

Pediatric asthma biomarker research has expanded from clinically accessible markers to diverse molecular and omics-based approaches. At present, FeNO and blood eosinophil count are the most clinically mature biomarkers, particularly for identifying type 2 inflammation and supporting selected treatment decisions. IgE sensitization, induced sputum, eosinophil granule proteins, and other candidate markers can provide additional phenotypic or mechanistic information, but their clinical utility remains context-dependent.

Most emerging and mechanistic biomarkers are not yet ready for routine pediatric asthma management. Their current value lies in clarifying disease biology, refining endotypes, and identifying future biomarker candidates. Clinical implementation will require age-appropriate validation, standardized assays, reproducible cutoffs, and evidence that biomarker-guided decisions improve outcomes beyond conventional clinical assessment, lung function, FeNO, blood eosinophils, and allergen sensitization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.alit.2026.08.003.

Conflict of interest
The authors have no conflict of interest to declare.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process
During the preparation of this manuscript, the authors used ChatGPT for language editing and improvement of readability. The authors reviewed and edited all AI-assisted content and take full responsibility for the final manuscript.

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